

Bluetooth in. Edge AI on board. Satellite out.

Firmware that listens to your **Bluetooth** devices, pre-processes with **Edge AI**, keeps what matters in flash, and delivers it over **LTE-M**, **NB-IoT** or **satellite NB-NTN**, choosing the cheapest uplink that works.

- ✓ Satellite NB-NTN registered, sent and acknowledged on a geostationary service
- ✓ Commissioned from a browser, with no cable and no app
- ✓ Signed updates over the air, over LTE-M and NB-IoT, that roll back unless they confirm

Built with  adsum
IOT CODER

THE HARDWARE · OUR PARTNER FANSTEL

An open gateway, made by Fanstel.

Fanstel designs, makes and sells the BLG20x and publishes its design: Bluetooth 6, cellular and satellite in a sealed IP67 enclosure, rated -40 to +80 °C. Fanstel is taking it through the hardware certifications for each region.



THE FIRMWARE · ADSUM NETWORKS

Industrial-grade IoT firmware.

Use it as is, or customise it in hours with Adsum IoT Coder. Every version is **certified on real hardware** by our reliability and security tests, and **kept secure** for as long as you renew.

From our firmware to your product, one choice at a time

Minutes

Try it free

Flash the demo image and judge it on your own desk.

Under 30 minutes

First gateway live

The ready-made image, commissioned and sending to your platform.

Under 2 hours

Make it yours

Customising with Adsum IoT Coder. Change the sensor side, or take the full source.

Then

Certify your version

Your build passes the same test suite, on the board.

Every year

Stay secure

Watched, advised and fixed. 12 months included, up to five years.

Three ways to have it. Ready-made image you change settings. Customisable image you change the sensor side; we keep the connectivity side signed. Full source you change anything. Every edition includes 12 months of security support and the evidence pack. Ask us for pricing.

Industrial grade, by design

- **Runs unattended.** Watchdogs and self-recovery from a hang, a dropped link, a bad configuration or a power cut.
- **Never loses data.** Records wait in flash until your platform acknowledges them. Alarms are never overwritten.
- **Updated over the air.** Signed updates over LTE-M and NB-IoT, so a security fix reaches the field with no site visit.
- **Edge AI, built for satellite.** Data is pre-processed on the gateway, so over satellite only alerts and status are sent.
- **Secure by design.** Signed boot, secure key storage, authenticated Bluetooth, and only the devices you name admitted.
- **Managed from afar.** Set up from a browser. Health reported, commands sent back, straight to your own platform.
- **Costs under control.** The cheapest uplink first, a data budget counted before every send, the modem off in between.
- **Tested to prove it.** Every version passes our reliability and security tests on the real board. See page 3.
- **Supported for years.** Security support for up to five years, with an evidence pack for your audits.

Applications

Remote assets and pipelines

Agriculture and livestock

Maritime, ports and buoys

Energy and utilities

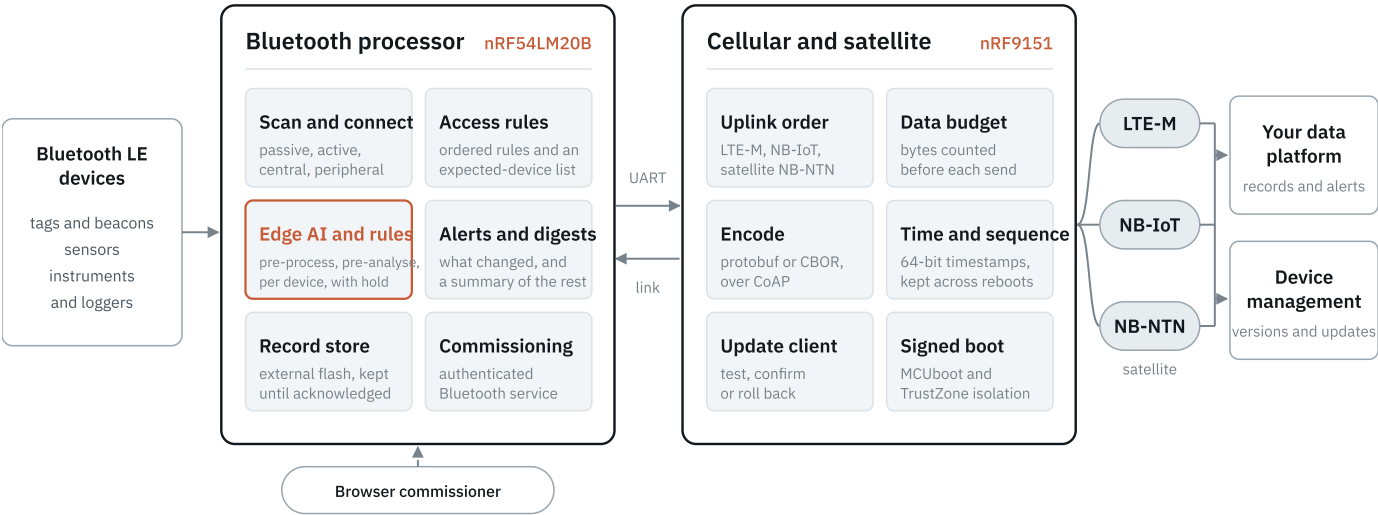
Cold chain and logistics

Environmental monitoring

HOW IT WORKS

Two processors, one gateway.

The Bluetooth processor listens, pre-processes and remembers. The cellular and satellite processor picks the uplink and delivers. One serial link joins them.



The access layer · yours to change. Delivered as source in the Customisable edition: which devices are admitted, and how they are checked, filtered and pre-processed. Signed with your key.

The connectivity core · signed by Adsum. Uplinks and their policy, transport security, secure boot and update, provisioning and metering, delivered as a signed image.

EDGE AI, AND WHY SATELLITE NEEDS IT

Send the decision, not the data.

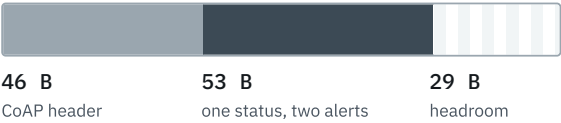
A satellite link carries a few kilobytes a month, not a stream. So the work happens on the gateway, on the nRF54LM20B beside the radio, before anything is sent.

- **Pre-process.** Filter and smooth each device’s readings, with its own state, thresholds and hold times.
- **Pre-analyse.** Detect a change, an anomaly or a missing device at the edge, and raise one alert instead of a thousand readings.
- **Prioritise.** Alerts go first, each with its age. The rest is summarised and waits for a cheaper uplink.
- **Hold.** Every record waits in flash until your platform acknowledges it, so nothing is lost while the sky is closed.

What is sent, by uplink

UPLINK	WHAT LEAVES THE GATEWAY	SIZE
LTE-M	Digests per device, alerts, status, a	~1 kB
NB-IoT	daily health record, and firmware updates	per cycle
Satellite	Alerts and status only. Everything else waits in flash for a cheaper uplink	128 B datagram

One satellite datagram 128 B maximum



A real datagram, sent and acknowledged over a geostationary NB-NTN service.

Uplink order: cheapest first, satellite last



Falls back	When an attach budget is spent or an uplink is removed, always with a stated reason.
Returns	To a cheaper uplink on a schedule, hourly by default.
Satellite needs	A position set at commissioning, NB-NTN modem firmware and a subscription.

SECURITY AND OPERATIONS

Updated over the air, over LTE-M and NB-IoT.

- **No site visit.** A security fix or a new feature reaches every gateway over the mobile network, while it keeps working.
 - **Safe by default.** Signed images only, for both processors. A new image is tested on the unit and rolls back by itself if it is not healthy.
 - **Never over satellite:** an update waits for a cellular uplink, so the satellite allowance is never spent on firmware.
- Measured on the hardware: 409 kB downloaded in 129 s, swapped, tested and confirmed.

INDUSTRIAL GRADE

Tested to prove it, on the real board.

Reliability: does it keep working?

Function	Every feature, end to end, on every release, across a library of Bluetooth devices, SIMs and networks.
Failover	Each uplink lost and recovered, LTE-M to NB-IoT to satellite and back; SIM, network and roaming refusals.
Stress	The full device list, floods of Bluetooth traffic, bursts of alerts, a full store, weak and patchy signal.
Durability	Soak runs for days; thousands of power cycles, cold boots and brown-outs; flash wear; memory watched for leaks.
Fault injection	Power cut mid-write and mid-update, a corrupted image, a bad configuration, a stalled processor. It must recover alone.
Updates	Every release applied over the air from the last one, interrupted, resumed, confirmed, and rolled back.

Security: can it be broken into?

Boot and update	Only signed images start. Tampered, unsigned and downgraded images are refused.
Hardening	Debug ports locked in production; no open shell, no default password, no secret left in the image; keys in secure storage.
Bluetooth	Pairing, replay and man-in-the-middle attacks; unauthenticated writes refused; a flood cannot fill the store.
Network	Encrypted transport checked; malformed, oversized, replayed and spoofed messages refused.
Fuzzing, pen test	Malformed data thrown at every input for hours; static analysis; a penetration test of each major release.
Supply chain	A software bill of materials (SBOM) from the real build, scanned against known vulnerabilities on every release.

Certified by Adsum

One exact version passed every test above, on the real board.

- **A certificate and a full test report** with every version.
- **Your version too:** your customised build earns its own certificate.
- **The hardware:** Fanstel is taking the BLG20x through the certifications for each region.

Security support

For every certified version you hold, for as long as you renew.

- **Watched and told:** new vulnerabilities, whether they apply, what to do meanwhile.
- **Fixed:** a security release, certified like any other.
- **Up to five years at once:** the support period the EU Cyber Resilience Act asks of a manufacturer.

The evidence pack

Your technical file for the firmware, kept complete for every version.

- Certificate and full test report
- SBOM in SPDX, advisory register and VEX
- Secure-by-design report, build record, changelog

Certified by Adsum means this version passed the Adsum Factory test suite on the board. It is not a conformity assessment under the Cyber Resilience Act or any other regulation. Radio type approval and electromagnetic compliance belong to the hardware and to Fanstel.

Firmware · Adsum Networks

Platform	nRF Connect SDK v3.4.0, Zephyr RTOS, MCUboot on both processors
Bluetooth LE	Passive and active scanning; three central sessions; LE Secure Connections only; 32 access rules and an expected-device list of 128
Uplink	LTE-M, NB-IoT, satellite NB-NTN; confirmable CoAP over UDP; protobuf or CBOR; records held in 2 MB flash until acknowledged
Updates	Over the air on LTE-M and NB-IoT, both processors; signed; test boot, health gate, automatic roll-back
Security	Signed boot; TrustZone isolation; bonded, authenticated commissioning from a browser

Hardware · Fanstel BLG20x

Cellular and satellite	Nordic nRF9151 SiP: LTE-M and NB-IoT (3GPP Release 14), NB-NTN (Release 17); Cortex-M33
Bluetooth and Edge AI	Nordic nRF54LM20B in a Fanstel BM20 module, Bluetooth 6.0 capable; Cortex-M33; with or without a power amplifier
Antenna, SIM, flash	One internal antenna for cellular and satellite; nano-SIM, eSIM footprint; 2 MB external flash
Wi-Fi 6, GNSS	Wi-Fi 6 (nRF7002) on some models, not used by this version; no GNSS, the position is set at commissioning
Power, enclosure	USB-C; IP67; -40 to +80 °C (maker's rating)
Certifications, sales	Regional hardware certifications in progress at Fanstel, who sells it: fanstel.com

Proven on the hardware. Satellite NB-NTN sent and acknowledged on a geostationary service; an LTE-M cycle delivered; NB-IoT attach on the nRF9151; an update downloaded over the mobile network, confirmed, and rolled back; a first run end to end, with a command back.

Notes for this version. The modem firmware is installed before deployment and is not updated in the field. Evaluation images carry a development key; a guided step provisions your own production key before the first volume flash, and Adsum never holds it. Radio, capacity and power figures are published with the confirmed shipping configuration.

Try it free, on your own desk.



support@adsumnetworks.com

www.adsumnetworks.com

The BLG20x, and the product brief

The hardware, from Fanstel



Install Adsum IoT Coder